

PARTH SHARMA

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SUMMARY

AI/ML Engineer (B.Tech, Artificial Intelligence & Data Science) with hands-on experience in **Python, machine learning, deep learning, GenAI, RAG, and FastAPI**, backed by a strong data foundation: SQL, ETL pipelines, and feature engineering on 500K+ records across banking, aviation, and academic domains. Experienced in building ML pipelines, comparing classification models, and exposing work through APIs, with a data-engineering internship that grounds models in clean, validated data.

EXPERIENCE

AI/ML Engineer Intern — Decode Data Academy July 2025 – Present

- Built data-preparation pipelines in **MySQL and SQL** to ingest, clean, and standardize 10,000+ records across 5+ branches and 2 colleges, eliminating 95%+ of data errors and producing **model-ready datasets** for analytics and ML use cases.
- Built 10–15 SQL views and 4–5 **star-schema** tables that make metrics and features easy to derive at scale.
- Automated recurring data processing with a **Pentaho Spoon** ETL layer, improving efficiency by 20–30%.
- Validated data quality and business rules across 1,000+ records so analysis and modeling run on trustworthy inputs.

PROJECTS

Insurance Incentive Optimization ([GitHub](#), [Live](#)) Python, FastAPI, Scikit-learn

- Built an end-to-end ML system that predicts **policy renewal probability** on 67,875 policyholder records and recommends the revenue-maximizing agent incentive (Rs. 500–4,500) per policy.
- Built a reusable **scikit-learn Pipeline** (median imputation, one-hot and target encoding, scaling, engineered **TotalLate** feature); compared Random Forest and XGBoost classifiers.
- Formulated the incentive search as an optimization problem with a diminishing-returns agent-effort model: $\text{net revenue} = \text{premium} \times (\text{renewal prob.} + \text{agent uplift}) - \text{incentive}$.
- Deployed a **FastAPI /predict** endpoint (Pydantic validation, CORS, batch input) on Render, consumed by an HTML/CSS/JS frontend live on Vercel.

Loan Default Risk Prediction ([GitHub](#)) Python, Pandas, Scikit-learn

- Built an end-to-end **ML pipeline** on a 55K-row, 122-column loan dataset: missing-value and outlier handling, encoding, and model-ready data preparation.
- Engineered **10+ predictive features** (income segmentation, credit ratio, EMI burden, employment duration) to improve customer risk profiling.
- Trained and compared **Logistic Regression, Decision Tree, and Gradient Boosting** classifiers to predict loan default risk; evaluated with standard classification metrics.

- Integrated **7 heterogeneous ride datasets** (465K+ raw records) into one schema; removed rate-card rows and 7K+ duplicates to produce 376K model-ready rows.
- Engineered cyclic (sin/cos) time features, location-frequency encodings for ~12.8K locations, and flagged/imputed distances; **detected and removed target-leaking columns** (e.g., weather, ratings).
- Built a leakage-free pipeline (train-only scalers, 70/15/15 split); trained HistGradientBoosting baselines and **PyTorch embedding MLPs** with early stopping and a 3-seed ensemble.

EDUCATION

- 2022 - 2026
- Bachelor of Technology (Artificial Intelligence and Data Science) at **Lakshmi Narain College of Technology, Bhopal, Madhya Pradesh, India**
- 2020 - 2021
- Higher Secondary Education (Class XII) at **Manikala Nahar Higher Secondary School**
- 2018 - 2019
- Secondary Education (Class X) at **Shri Maniklal Nahar International School**

TECHNICAL SKILLS

Programming:	Python, SQL, DAX
Machine Learning:	Classification (Logistic Regression, Decision Trees, Gradient Boosting), Feature Engineering, Model Evaluation
Deep Learning & GenAI:	Deep Learning, Generative AI, LLMs, Retrieval-Augmented Generation (RAG)
APIs & Deployment:	FastAPI, REST APIs, Model Deployment
Libraries:	Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, Matplotlib, Seaborn
Data Engineering:	ETL Pipelines, Pentaho Spoon, Data Validation, Star Schema
Databases:	MySQL, Relational Databases
Tools:	Git, Jupyter Notebook, Power BI, Microsoft Excel
Coursework:	Machine Learning, DBMS, Statistics, Data Structures and Algorithms

CERTIFICATIONS

- AWS Academy Data Engineering Badge

Dec 2025
- Wipro TalentNext Database Solutions Expert

Oct 2024
- HackerRank SQL – 5-Star (Gold) Badge